

NETWORKS AND COMPLEXITY

Solution 12-3

*This is an example solution from the forthcoming book *Networks and Complexity*.*

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 12.3: Infinite excess degree [Lvl. 1]

Consider a network with $p_0 = 0$ and $p_k = ak^{-3}$ for $k > 0$. Show that this network has finite mean degree but infinite mean excess degree. You can use the $\sum_{k=1}^{\infty} k^{-2}$ is finite while $\sum_{k=1}^{\infty} k^{-1}$ diverges. [Hint: Don't try to determine a .]

Solution

The mean degree of our network is

$$z = \sum_{k=1}^{\infty} kp_k \quad (1)$$

$$= \sum_{k=1}^{\infty} k ak^{-3} \quad (2)$$

$$= a \sum_{k=1}^{\infty} k^{-2} \quad (3)$$

which is finite. We now compute the excess degree distribution

$$q_k = \frac{(k+1)p_{k+1}}{z} \quad (4)$$

$$= \frac{a}{z}(k+1)(k+1)^{-3} = \frac{a}{z}(k+1)^{-2}. \quad (5)$$

Note that we can use this equation for all k including 0. In the next step we want to show that the mean excess degree is infinite, so our strategy is to create an expression of the form

$\sum_{k=1}^{\infty} k^{-1}$. We compute

$$q = \sum k q_k \tag{6}$$

$$= \sum \frac{a}{z} k (k+1)^{-2} \tag{7}$$

$$= \sum \frac{a}{z} (k+1) (k+1)^{-2} - \frac{a}{z} (1) (k+1)^{-2} \tag{8}$$

$$= \left(\sum \frac{a}{z} (k+1) (k+1)^{-2} \right) - \left(\sum \frac{a}{z} (k+1)^{-2} \right) \tag{9}$$

$$= \frac{a}{z} \left(\sum (k+1)^{-1} \right) - \frac{1}{z} \left(\sum a (k+1)^{-2} \right) \tag{10}$$

$$= \frac{a}{z} \left(\sum_{k=1}^{\infty} k^{-1} \right) - \frac{1}{z} \left(\sum_{k=1}^{\infty} a k^{-2} \right) \tag{11}$$

$$= \frac{a}{z} \left(\sum_{k=1}^{\infty} k^{-1} \right) - \frac{z}{z} \tag{12}$$

$$= \frac{a}{z} \left(\sum_{k=1}^{\infty} k^{-1} \right) - 1 \tag{13}$$

$$= \infty - 1 = \infty \tag{14}$$

$$\tag{15}$$